General Structure of the Model Solution

Pascal Michaillat
https://pascalmichaillat.org/w/

Solution to the macro matching model

Need to solve a 2x2 system

2 variables : y , x

2 equations : aggregate demand, aggregate supply

The solution is given by

$$\begin{cases} y = \sigma(x)\left[\underline{f(x)\,k} + \mu/p\right] \\ y = f(x)\cdot k \end{cases}$$

$\sigma/1-\sigma = \dfrac{MPS}{1-MPS}$

endowment of wealth

AS curve. $y^s(x) = f(x)\cdot k$

AD curve (pure) $y^d(x) = \dfrac{\sigma(x)}{1-\sigma(x)}\cdot\dfrac{\mu}{p} = \dfrac{x^{\varepsilon}}{[1+\tau(x)]^{\varepsilon-1}}\,\dfrac{\mu}{p}$

Behavioral curve: (behavioral) $y^b(x) = \sigma(x)\left[y^s(x) + \dfrac{\mu}{p}\right]$

$$\boxed{y^b(x) = \sigma(x)\,y^s(x) + [1-\sigma(x)]\,y^d(x)}$$

Behavior of household is linear combination of supply & demand b/c $\begin{cases} \sigma \in (0,1) \\ 1-\sigma \in (0,1) \end{cases}$

Spending

Two equivalent formulation of the solution. (RA / HA)

$$\begin{cases} y = y^b(x) \\ y = y^S(x) \end{cases}$$

$$\Leftrightarrow \begin{cases} y = \sigma(x)\, y^S(x) + [1-\sigma(x)]\, y^d(x) \\ y = y^S(x) \end{cases}$$

$\in (0,1)$

$$\Leftrightarrow \begin{cases} y = y^S(x) \\ y^S(x) = \underline{\sigma(x)\, y^S(x)} + \underbrace{[1-\sigma(x)]}_{\in(0,1)}\, y^d(x) \end{cases}$$

$$\Leftrightarrow \begin{cases} y = y^S(x) & \leftarrow \text{output given by AS} \\ y^S(x) = y^d(x) & \leftarrow \text{tightness at AS = AD} \end{cases}$$